

Applied Deep Learning

Chapter 1: Preliminaries

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Winter 2026

Deep Learning More than Ever

Deep learning is turning into an *unbeatable champion*

- It has achieved accuracy levels in image classification, restoration, and segmentation that *surpass human capabilities*
- It can create *human-like conversational dialogues* and respond to scientific and technical questions
- It can *beat world-class* chess players
- It has solved problems that *had not been* effectively addressed for a *long time*

We all have items to add to this list!

The Unbeatable: AlexNet $\rightsquigarrow$ ResNet

Image recognition has been one of the first problems in AI

we want the machine to recognize what in an image is

- Yann LeCun proposed LeNet in 1998
- In 2010, ImageNet started the annual contest

ImageNet Large Scale Visual Recognition Challenge

↳ AlexNet (2012) $\rightsquigarrow$ VGG and GoogLeNet (2014) $\rightsquigarrow$ ResNet (2015)

- Currently, we have much deeper and advanced neural networks

We will get what these networks are and do in this course

The Unbeatable: AlexNet $\rightsquigarrow$ ResNet

In February 2015, first Microsoft and then Google announced that their deep neural networks beat human accuracy in image recognition

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Microsoft's Deep Learning Project Outperforms Humans In Image Recognition

Michael Thomsen Former Contributor @

I write about tech, video games, science and culture.



Feb 19, 2015, 01:06pm EST

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Microsoft, Google Beat Humans at Image Recognition

By R. Colin Johnson 02.18.2015 0

The Unbeatable: ChatGPT

Making a *human-like dialogues* had been another fundamental problem in AI

- ↳ ChatGPT broke the record of fastest-growing software in history
- ↳ We are working with GPT 5.1 today!



The Unbeatable: *AlphaZero*

Computers playing chess: a long-going goal in AI

- 1951: Alan Turing designed a program that plays chess
- 1989: Gary Kasparov defeated IBM's Deep Thought
- 1996: Gary Kasparov defeated IBM's Deep Blue
- 1997: IBM's Deep Blue defeated Gary Kasparov
- 2005: Ruslan Ponomarev was the last person beating AI chess player
- 2017: DeepMind released AlphaZero algorithm
 - ↳ AlphaGo $\rightsquigarrow$ AlphaGo Zero $\rightsquigarrow$ AlphaZero
 - ↳ After 24 hours of training it defeated Stockfish chess engine
- Even newer AI chess players, e.g., Leela Chess Zero (2019)

The Unbeatable: *Neural Machine Translation*

Most *translator machines* were unreliable until *AI kicked in!*

- Computer-based translation is a research topic *since 1950s*
- In 2000s, *statistical machine translation (SMT)* became dominant
- In 2010s, *neural machine translation (NMT)* showed *great potentials*
 - ↳ *Baidu launched first large scale NMT in 2015*
 - ↳ *Google introduced its NMT called GNMT in 2016*
 - ↳ *DeepL was launched in 2017*
- *And again: lots of NMT nowadays are getting developed!*

Notion of Deep Learning

Glum starts this conversation

- + *Well! Nice examples! But, what exactly is this Deep Learning?*
- Deep Learning refers to the subset of Machine Learning that uses Deep Neural Networks to execute a given learning task
- + *Wait a moment! The only thing that I understood is that **Deep Learning is a form of Machine Learning!** But,*
 - *what is Machine Learning itself?*
 - *what are Neural Networks in the first place? Before we talk about their “deep” version*
 - *what do you mean by a “learning task”?*
- We are going to understand all of them!
 - First, we start with the concept of Machine Learning